```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
dataset = sns.load_dataset('titanic')
dataset.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Next steps: Generate code with dataset New interactive sheet

```
dataset.shape
```

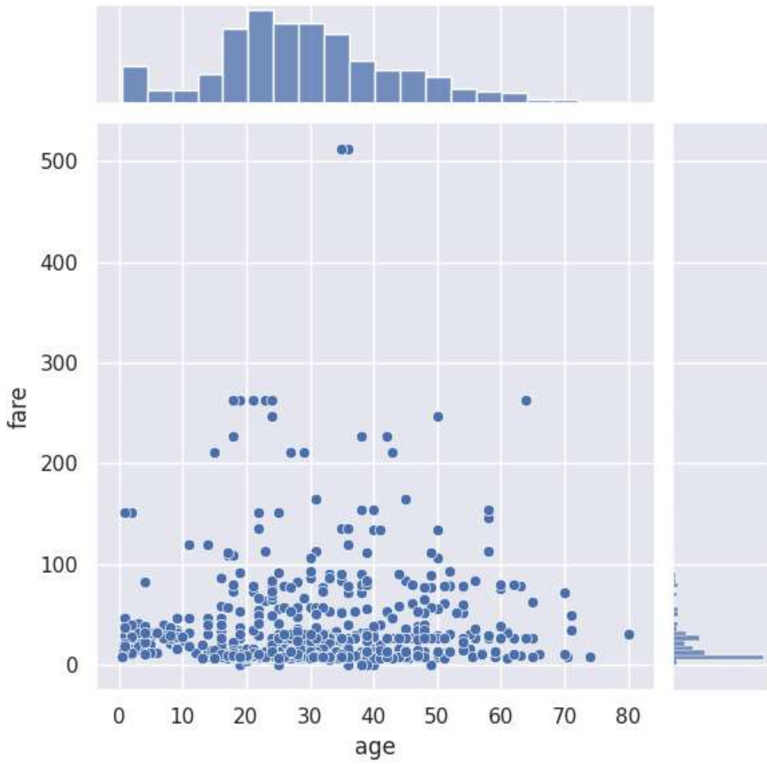
```
(891, 15)
```

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   survived     891 non-null    int64   
 1   pclass       891 non-null    int64   
 2   sex          891 non-null    object  
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    object  
 8   class        891 non-null    category
 9   who          891 non-null    object  
 10  adult_male   891 non-null    bool    
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object  
 13  alive        891 non-null    object  
 14  alone        891 non-null    bool    
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

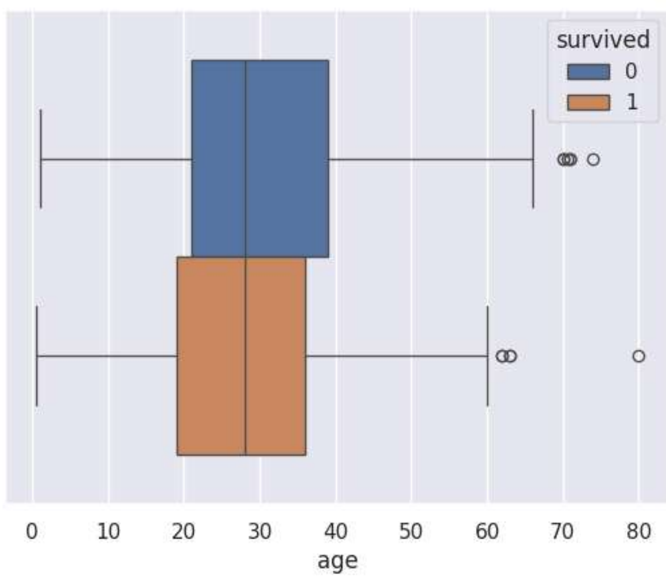
```
sns.jointplot(x='age', y='fare', data=dataset)
```

```
<seaborn.axisgrid.JointGrid at 0x7898499fe870>
```




```
sns.boxplot(x='age', data=dataset, hue="survived")
```

```
<Axes: xlabel='age'>
```




```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
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 2   sex          891 non-null    object  
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    object  
 8   class        891 non-null    category
 9   who          891 non-null    object  
```
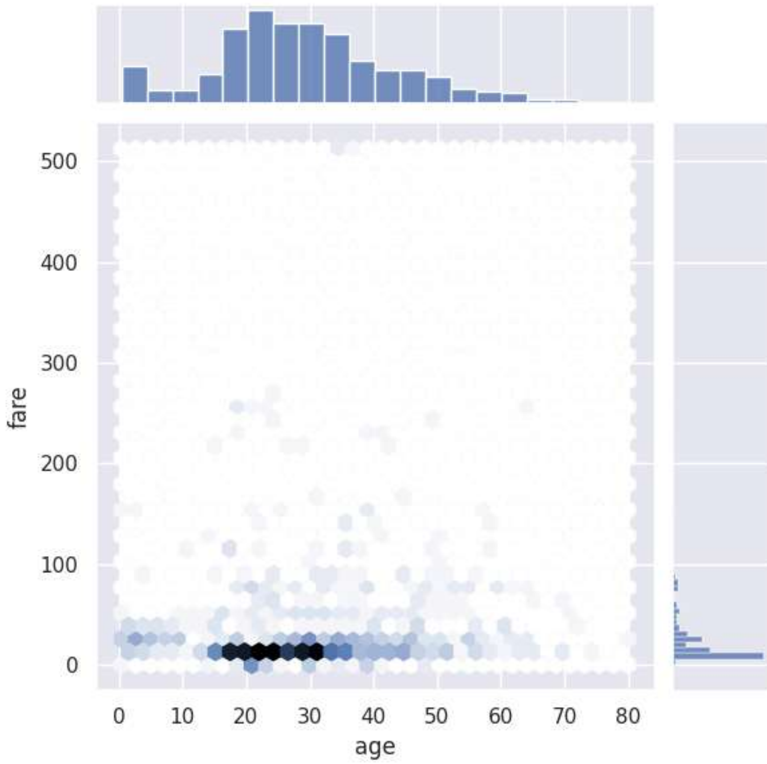

```
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

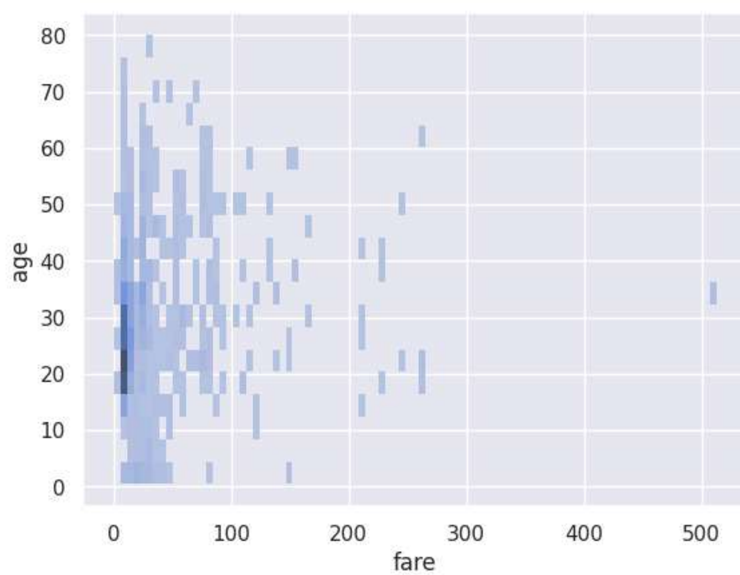
```
sns.jointplot(x='age', y='fare', data=dataset, kind='hex')
```

```
<seaborn.axisgrid.JointGrid at 0x789849a22120>
```




```
sns.histplot(data=dataset,x="fare",y="age")
```

```
<Axes: xlabel='fare', ylabel='age'>
```




```
x=dataset["fare"]
x
```

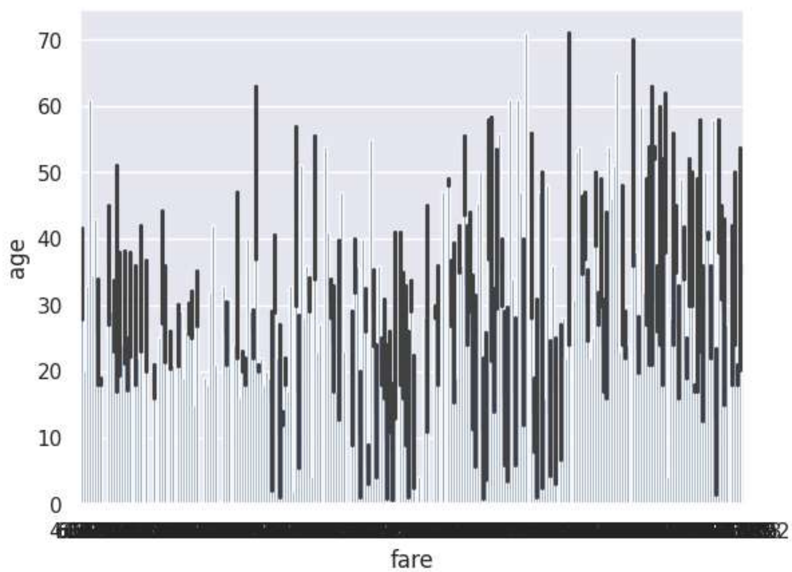
	fare
0	7.2500
1	71.2833
2	7.9250
3	53.1000
4	8.0500
...	...
886	13.0000
887	30.0000
888	23.4500
889	30.0000
890	7.7500

891 rows × 1 columns

dtype: float64

```
sns.barplot(data=dataset,x="fare",y="age")
```

```
<Axes: xlabel='fare', ylabel='age'>
```

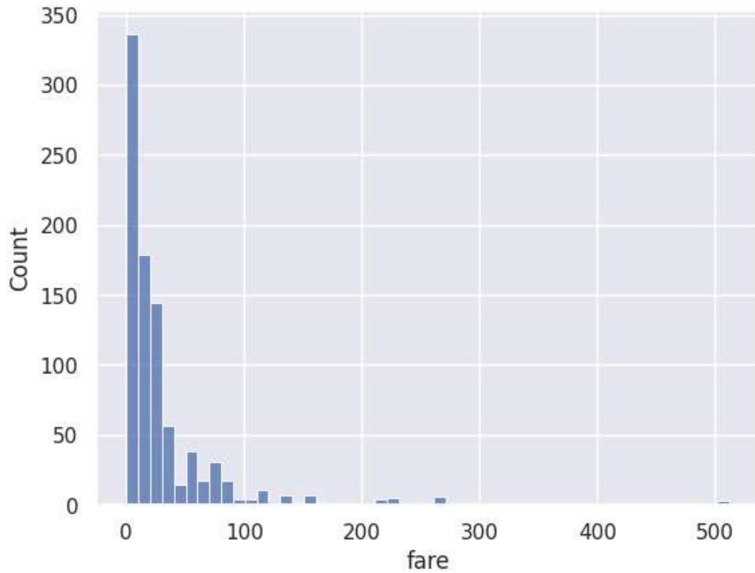



```
sns.pairplot(dataset)
```

```
<seaborn.axisgrid.PairGrid at 0x789849409760>
```

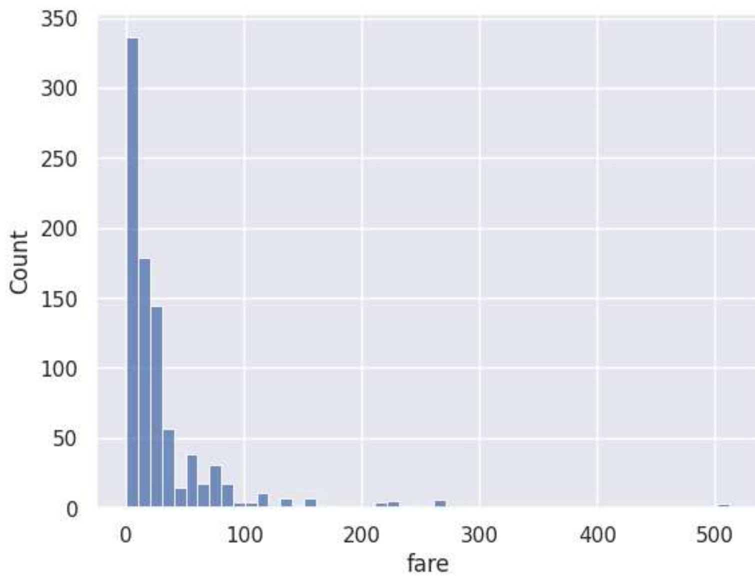

```
sns.histplot(data=dataset,x="fare",binwidth=10)
```

```
<Axes: xlabel='fare', ylabel='Count'>
```




```
sns.histplot(data=dataset,x="fare",binwidth=10)
```

```
<Axes: xlabel='fare', ylabel='Count'>
```




```
sns.histplot(data=dataset,x="fare",binwidth=30)
```

```
<Axes: xlabel='fare', ylabel='Count'>
```

```
import numpy as np
dataset.select_dtypes(include=np.number).corr(method='pearson')
```

	survived	pclass	age	sibsp	parch	fare
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000

```
dataset.loc[:,["survived","alive"]]
```

	survived	alive
0	0	no
1	1	yes
2	1	yes
3	1	yes
4	0	no
...	...	...
886	0	no
887	1	yes
888	0	no
889	1	yes
890	0	no

891 rows × 2 columns

```
dataset_cleaned=dataset.drop(['pclass','embarked','deck','embark_town'],axis=1)
```

```
sns.boxplot(data=dataset_cleaned,x="sex",y="age", color="Purple")
```

```
<Axes: xlabel='sex', ylabel='age'>
```